User Acceptance Testing (UAT) Master Plan — WariMitra

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Classification: UAT Strategy, Governance & Execution Framework

Standard: IEEE Std 829-2008 / BABOK v3 Chapter 8 (Solution Evaluation)

UAT Lead: Senior Technical Business Analyst / Product Owner

Testing Environment: Firebase Staging Project (`asia-south1`) with Flutter Mobile Staging Build

Source Evidence: `IEEE_TEST_CASES.md:1-694`, `TESTING_GUIDE.md`, `functions/src/index.ts:336-419`

1. Role of the Business Analyst in UAT

In enterprise and GovTech software delivery, the Business Analyst serves as the **bridge between technical QA and real-world operational stakeholders**. While QA validates that code meets technical specifications without syntax crashes, the BA ensures that the software **delivers business value, satisfies statutory mandates, and functions under extreme field conditions**.

Specific BA Responsibilities During UAT:

1. **Scenario Design & Day-in-the-Life Scripts:** Translating technical user stories into contextual field simulations (e.g., simulating a medical emergency during a 40°C afternoon halt).
2. **Business User Facilitation:** Guiding rural Dindi leaders, police inspectors, and medical officers who may have low digital literacy through test scenarios in Marathi.
3. **Defect Triage & Severity Classification:** Evaluating whether a reported bug is a cosmetic glitch, a functional defect, or a critical life-safety blocker.
4. **Acceptance Criteria Validation:** Verifying that actual system outputs match the BRD and FRD Given-When-Then definitions.
5. **Business Sign-Off Facilitation:** Securing formal sign-off from District Collectors, Police Superintendents, and Chief Medical Officers prior to production deployment.

2. UAT Scope & Test Strategy

2.1 In-Scope Test Tracks

- **Track 1: Pilgrim Field Usability:** Mobile app navigation in Marathi, 3-second hold SOS gesture, offline cached map reading, family circle invite code joining.
- **Track 2: Emergency Response Triage:** Web Command Center incident receipt, real-time alert updates without page refresh, status transitions (`pending` $\rightarrow$ `resolved`).
- **Track 3: Healthcare Camp Operations:** Geohash medical camp discovery, distance sorting, bed capacity updates.
- **Track 4: Missing Persons & Lost/Found:** Photo upload, case status updates, and public notification.
- **Track 5: Governance & Web3 Audit:** Verification of SHA-256 hash generation and transaction proof visibility on the dashboard.

2.2 Testing Environments & Infrastructure

- **Backend:** Firebase Project Staging (`warimitra-staging`), Cloud Functions deployed in `asia-south1`.
- **Mobile Clients:** Android APK builds distributed to 50 field test devices (Samsung Galaxy A03 / Redmi 9A entry-level hardware to simulate real pilgrim phones).
- **Web Command Center:** Next.js staging build accessed via Chrome/Firefox on police control room terminals.
- **Test Case Management:** Synchronized via the implemented test results Cloud Functions (`saveTestCaseResults` / `getTestCaseResults` writing to `testResults/shared`) (`functions/src/index.ts:342-419`).

3. Entry & Exit Criteria

3.1 UAT Entry Criteria (Must be met before UAT starts)

1. All System Integration Testing (SIT) test cases executed with $\ge 95\%$ pass rate.
2. Zero open Critical (Severity 1) or High (Severity 2) technical defects in Jira/Firestore.
3. Firebase Staging environment provisioned with realistic seed data (50 medical camps, 20 temple queues, 100 sample users).
4. Bilingual UI copy (Marathi / English) fully populated with zero placeholder strings.

3.2 UAT Exit Criteria (Required for Production Go-Live)

1. 100% of Critical Life-Safety UAT test cases (SOS trigger, rate limiting, geofencing) passed.
2. Minimum 90% pass rate across secondary operational workflows.
3. Formal digital sign-off obtained from Lead Police, Medical, and District Administrative stakeholders.
4. Comprehensive defect log signed off with all residual cosmetic issues deferred to Post-Wari v2.1.

4. Defect Severity Matrix & Triage Protocol

Severity Level	Operational Impact	SLA for Fix	Example in WariMitra
S1 — Critical Blocker	Prevents emergency SOS dispatch; crashes web dashboard; drops patient records; database rules expose private data.	≤ 4 Hours	Hold SOS button fails to invoke `createSOS`; rate limiter permanently locks out valid users.
S2 — Major Business Defect	Core workflow impaired; reasonable workaround exists but causes operational friction.	≤ 24 Hours	Medical camp distance sorting returns incorrect kilometer values; Marathi text overlaps on small screens.
S3 — Minor Defect	Non-critical feature defect; cosmetic issue; minor formatting error.	Next Sprint / Post-Wari	Icon misalignment on Lost & Found card; minor delay in Web3 proof badge rendering.
S4 — Enhancement Request	User suggests improvement not defined in baseline BRD.	Backlog Review	Request to add Marathi voice reading of temple queue wait times.